Observability versus monitoring

Observable, observer, observability

- **Observable:** data that can be monitored.
- **Observer:** persons or tools that observe data.
- **Observability:** the ability to observe data from applications with tools.
- **Continuous monitoring:** process and technology used to detect compliance and risk issues associated with an organization's financial and operational environment.
- **Continuous observability:** enables DevSecOps teams to observe application telemetry and find runtime security compliance, and privacy risks in the pipeline.



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On-demand observability

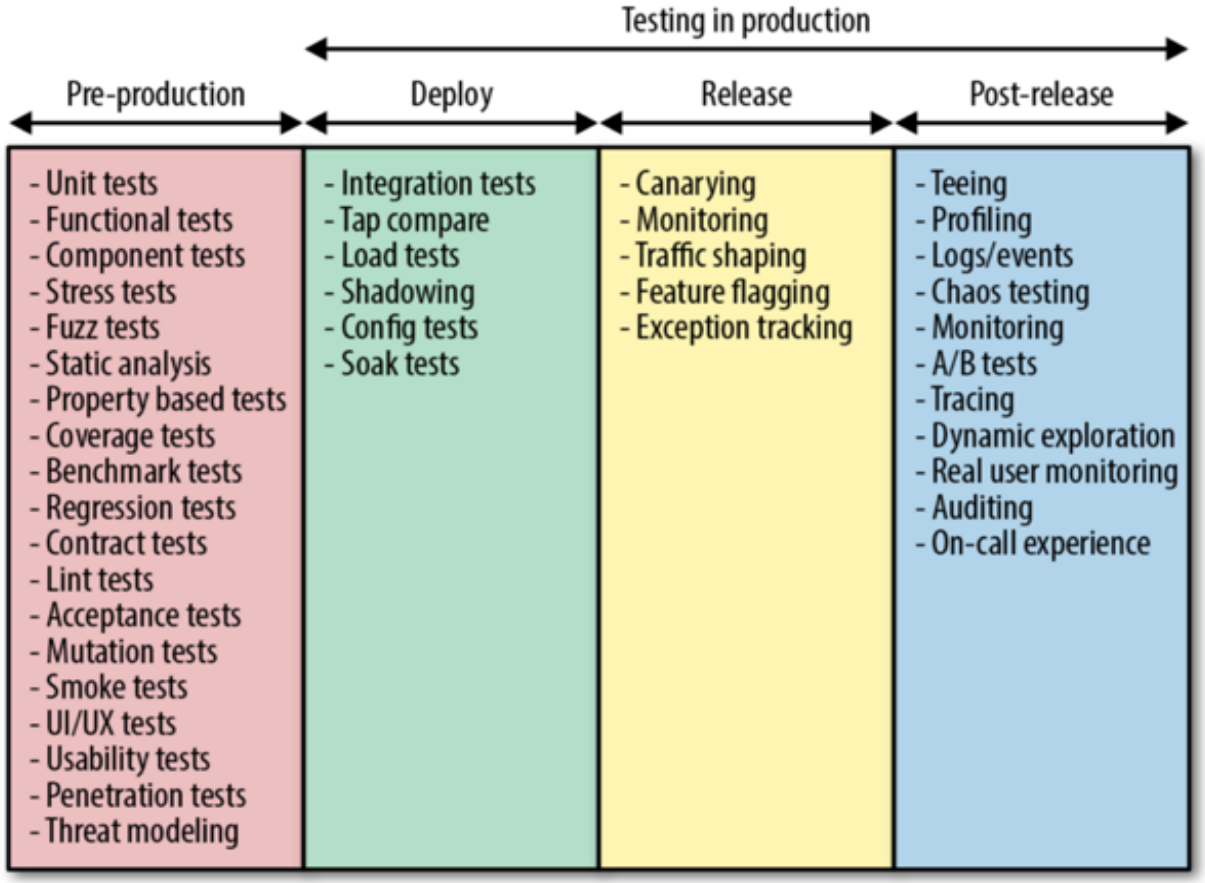
Testing is not sufficient:

- Observability is needed to correctly determine test verdicts.
- Observability is needed to support debugging.

Determine system features needed to get data for observability.

Build observable systems:

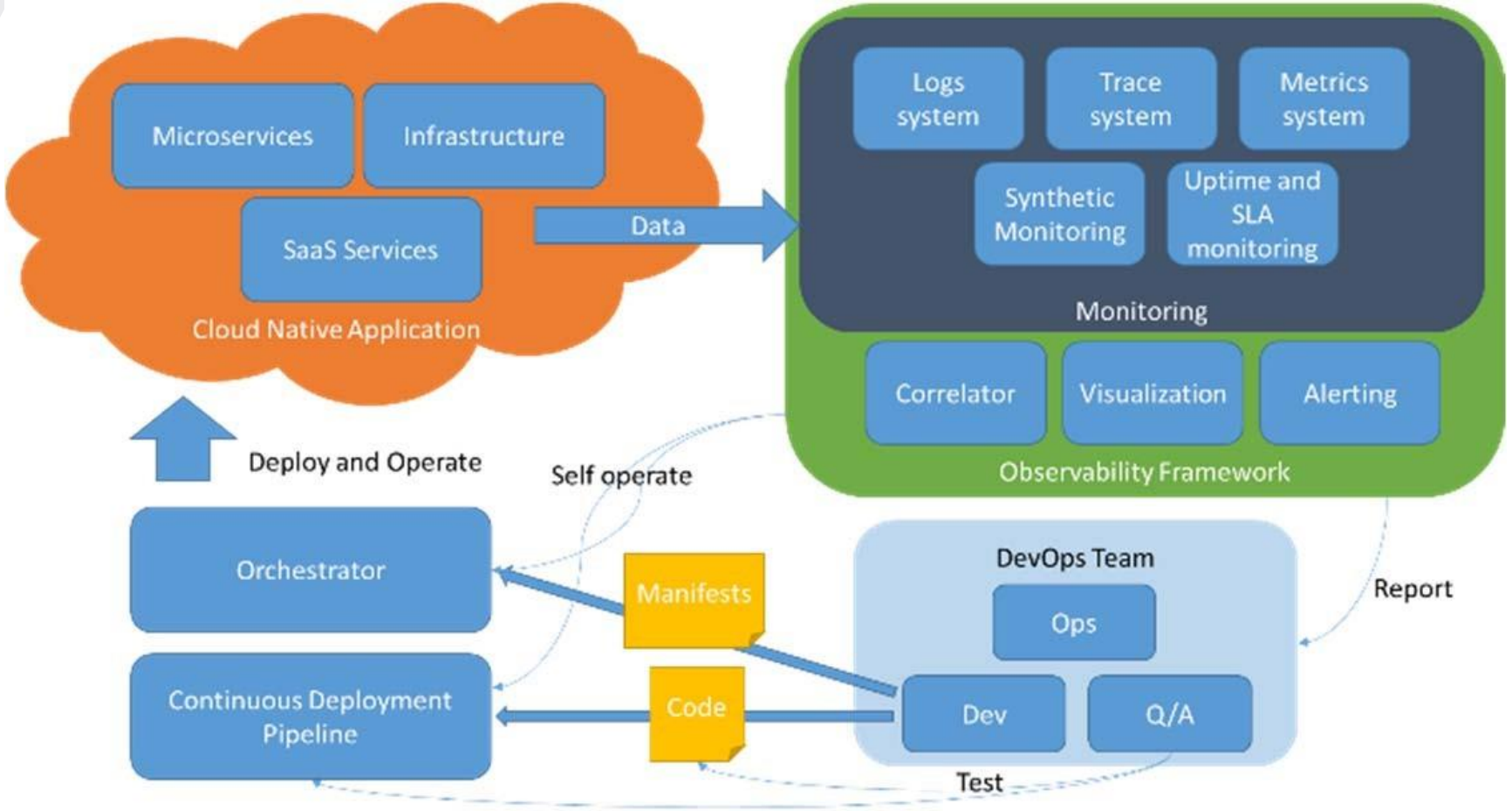
- Behaviour observability should be primary feature.
- Provide multiple windows into functions.



Source: https://sookocheff.com/post/architecture/testing-in-production/

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Continuous observability



Source: https://www.semanticscholar.org/paper/Demonstration-of-an-Observability-Framework-for-Marie-Magdelaine-Ahmed/612f8eb9de65aa4c592d87f44c4aad4ab18df883

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SWIFT hacking

- North Korea seeks to evade sanctions.
- Targeted Society for Worldwide Interbank Financial Telecommunication (SWIFT)
- Central Bank of Bangladesh banks with Federal Reserve Bank of New York:
  - February 4th, 2016; $1,000,000,000 in 13 transactions.
  - CBB printer and logs failed.
  - Analysts in New York stopped over $850,000,000 in transfers.
- Observability matters, in multiple places.
- IaC allows multiple metrics.



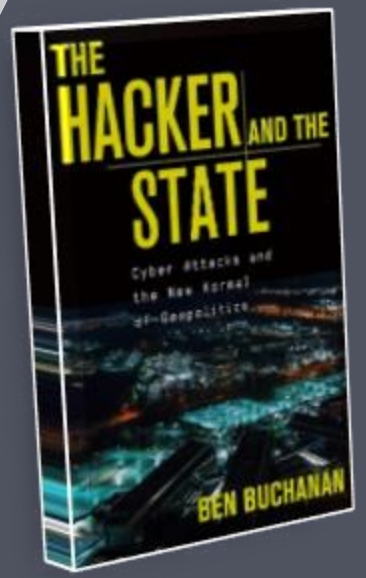
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Case story: The Hacker and the State | Takeaways

Ben Buncham,
Author of *The Hacker and the State: Cyber Attacks and the New Normal of Geopolitics*



- Every case of hacking involves someone observing the wrong thing, or the right thing at the wrong time.
- Buchanan contains source data for the earlier SWIFT story.



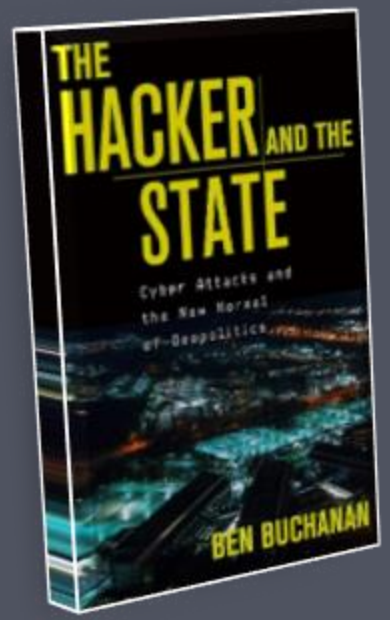
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Case story: The Hacker and the State | Benefits

Ben Buncham,
Author of *The Hacker and the State: Cyber Attacks and the New Normal of Geopolitics*



- DevOps observability requires knowing how operations work and who might break them.
- Examples of previous failures allow growth and expansion.

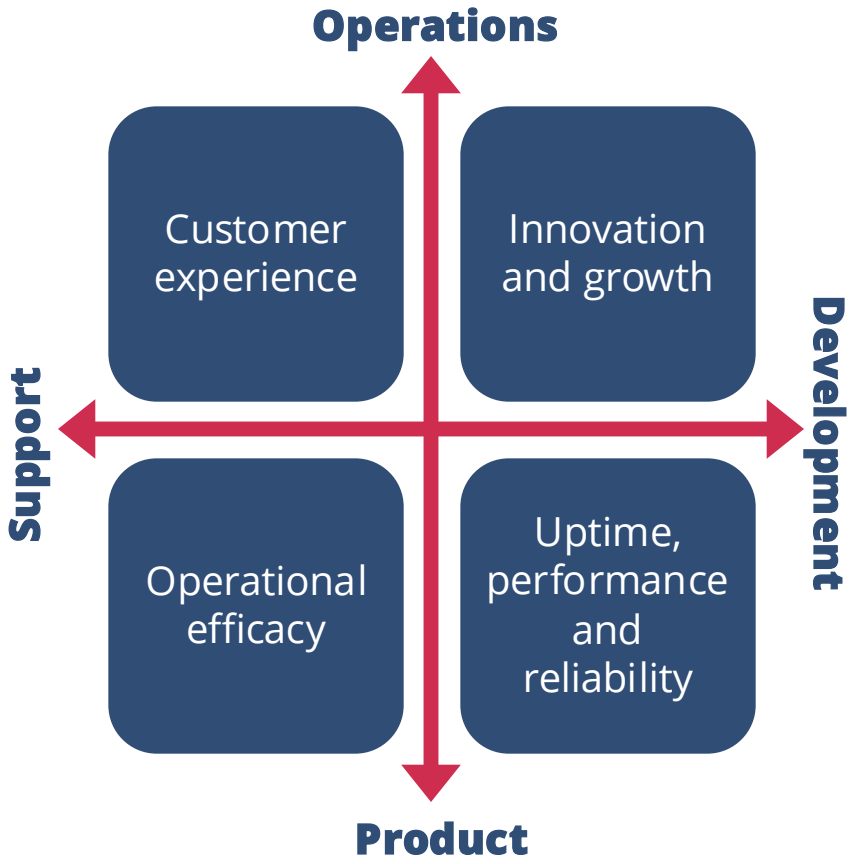


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Observable maturity

Track experience indicators
Quality foundation
Bottom-of-the-funnel analysis (BotFA)

Service characterization
Resolve dependency
Risk
Automate
Instrumentation
Resource sizing



Measure code pipelines
Iterate and measure impact
Release Quality Management (RQM)

Alert Quality Management (AQM)
Incident retrospectives
Incident orchestration
Service Level Management (SLM)

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